

# Meeting Summaries of September 5th 2025

**Date:** September 5th, 2025

**Time:** 19:00–20:00 PM(PCT)

**Attendees:** Prof.Jan, Prof.Jabed, Prof.Yue and Feng

**Topic:** Feng's presentation of the project framework and feedbacks about modifications on the proposal

## Summary of Discussion

- Feng presented the framework of the project, majorly about the three interacting components, chemical, taxa and environmental, and the project objectives: develop a taxa indicator to infer the probability of a site being degraded.
- Feng discussed the idea of controlling the variation that was contributed by environmental factors to the community composition: an environment-reference community ruler(as shown in Figure 1) was proposed to measure how the counterfactual would be if there was no pollution given the environmental conditions.
- Feng talked about the steps to construct and apply this ruler and how to connect the difference between the observed community composition and the readout composition to the pollution level, where the regression problem(piecewise quantile regression) would come in.
- Prof. Jabed and Prof. Jan asked several questions about the construction and application of the ruler, helped Feng to clarify the ideas and gave some suggestions about the proposal writing.

## Decisions Made

### 0.1 About words and expressions in the presentation and proposal

- Some choice of words and expressions in giving a presentation can be improved, for example, using "this 'symbol' will be explained later" can leave some space for notations coming later and make the communication more smooth.
- Some professional terms need to be used and explained carefully, for example, "community composition" should be used instead of "community structure", concise definition of 'reference site' is also necessary as a foundation of the analysis.

### 0.2 About the proposal writing

- The proposal can be revised with **more clearly explained words and discussions** so that a reader without background of ecology and a little about statistics can understand the objectives and workframe of the project.
- **The hierarchy of the proposal can be improved:** the existing workframe from Jian can be one section and the new coming analysis from my work can be another section. This will make the proposal and the following work organized in distinguishing existing workframe and new contributions.

- There need some citations to support the talk in 'Introduction' section, for example, these factual statements and the realistic necessity for solving the problems.

### 0.3 About the three major new contributions

Discussed and advised by professors, there are three major new contributions in Feng's project compared to Jian's work:

- Explore the **existence and location of breakpoints with Piecewise Quantile Regression** in the numerical relationship between pollution level and the shifts of community composition.
- Discuss the potential usefulness of **spatial information** and include it to develop the analysis with spatial relevant methods.
- Investigate the effect of **sample size** on the building of the taxa indicator and the inference accuracy, come up with a criteria for the minimum sample size needed for this and similar studies.

### Action Items

- Prof. Jan would provide(have provided) some references to help Feng understand and conduct the discussed improvements above.
- Feng will go to modify and rearrange some proposal sections, including Introduction, distinguishing existing workframe and new contributions, and refining the words and expressions for clarification.
- Feng will keep reading the articles and organize the current work based on the insights gained.

### Follow-Up

Temporary meeting time: September 11th, 2025, time to determine between 14:00 to 18:00 PM(PCT), lasting for about 40 minutes.

### Supportive materials

- A meeting recording was stored in Feng's computer.
- The presentation slide have been uploaded to the Google Drive folder - 'Sep\_4th\_25\_meeting' in the 'Thesis\_Meeting\_Notes' directory. [Click here](#) to the Google Drive - 'Thesis\_Meeting\_Notes'

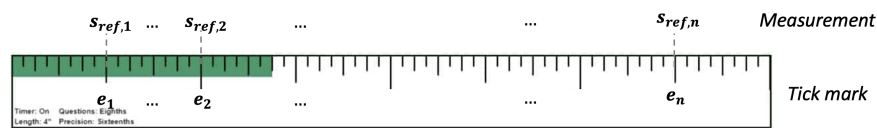
## 1 Appendix

### 1.1 Figures

To a site  $i$ , its environmental condition and reference community composition are  $e_i, s_{ref,i}$ :

$$e_i \in \{e_1, \dots, e_n\}$$

$$s_{ref,j} \in \{s_{ref,1}, \dots, s_{ref,m}\}$$



**An ruler of environmental condition for perfect community composition (conterfactual)**

Figure 1: Idea of the environment reference community composition ruler